q5_prime.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

unsigned long checks = 0;

int is_prime_naive(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    for (unsigned long i = 2; i < n; i++) { checks++; if (n % i == 0) return 0; }
    return 1;
}

int is_prime(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    for (unsigned long long i = 2; i * i <= n; i++) {
        checks++;
        if (n % i == 0) return 0;
    }
    return 1;
}

int is_prime_while(unsigned long n)
{
    checks = 0;
    if (n < 2) return 0;
    unsigned long long i = 2;
    while (i * i <= n) {
        checks++;
        if (n % i == 0) return 0;
        i++;
    }
    return 1;
}

int is_prime_dowhile(unsigned long n)
{
    checks = 0;
    if (n < 4) return n >= 2;
    unsigned long long i = 2;
    do {
        checks++;
        if (n % i == 0) return 0;
        i++;
    } while (i * i <= n);
    return 1;
}

int main(void)
{
    unsigned long tickets[] = {97, 100, 7919, 1000003};
    for (int k = 0; k < 4; k++) {
        int p = is_prime(tickets[k]);
        printf("%8lu %-9s divisions = %lu\n", tickets[k], p ? "prime" : "not prime", checks);
    }
    printf("\nComparison of divisions (naive / for / while / do-while):\n");
    for (int k = 0; k < 4; k++) {
        unsigned long n = tickets[k], a, b, c, d;
        is_prime_naive(n); a = checks;
        is_prime(n); b = checks;
        is_prime_while(n); c = checks;
        is_prime_dowhile(n); d = checks;
        printf("%8lu  %8lu  %6lu  %6lu  %6lu\n", n, a, b, c, d);
    }
    return 0;
}
```

```
Terminal  -  output
```

```
user@pc:~/Assignment2$ gcc -Wall q5_prime.c -o q5 && ./q5
      97 prime     divisions = 8
     100 not prime divisions = 1
    7919 prime     divisions = 87
 1000003 prime     divisions = 999

Comparison of divisions (naive / for / while / do-while):
      97        95       8       8       8
     100         1       1       1       1
    7919      7917      87      87      87
 1000003   1000001     999     999     999
user@pc:~/Assignment2$ _
```